

Statement of Purpose

My interest in Computer Science has grown through building things rather than only studying concepts. As a Computer Science and Engineering undergraduate at VNR VJIET, I have gradually moved from learning programming fundamentals to developing web applications, solving algorithmic problems, and exploring Artificial Intelligence. These experiences have helped me understand that computer science is not limited to writing code. It involves understanding problems deeply, designing efficient solutions, and continuously improving them. I now want to pursue a Master's degree in Computer Science to strengthen my theoretical foundation, gain research experience, and work on problems that require deeper technical understanding.

My academic journey at VNR VJIET has given me a strong foundation in programming, Data Structures and Algorithms, Object-Oriented Programming, databases, operating systems, and web development. I currently maintain a CGPA of 8.93/10. My academic performance has also been consistent from an early stage. I was recognized as a topper during my school education and received academic recognition and scholarships for my performance. Although these achievements are important to me, the more significant outcome has been developing the discipline to learn consistently and work towards measurable goals.

As I progressed through my undergraduate studies, I became more interested in applying what I learned to practical systems. One of my major projects was a Library Management System developed for the CSE department environment. I worked with JavaScript, Node.js, Express.js, MongoDB, and REST APIs to build role-based functionality for administrators, faculty, and students. The system included book issue and return transactions, automatic fine calculation, notification support, and a module through which students could communicate knowledge requests to faculty. Working on this project taught me that creating a useful application requires much more than implementing individual features. I had to think about data organization, user roles, API design, reliability, and how different components work together.

I also developed a Habit Tracker application, where I worked on recurrence schedules, streak calculation, database storage, and data visualization. Building the application gave me practical exposure to designing a system where the frontend, backend, and database continuously interact with each other. These projects strengthened my interest in software engineering and made me curious about how larger and more complex systems are designed to remain efficient, scalable, and reliable.

My interest in Artificial Intelligence developed alongside my software development experience. During my Agentic AI industrial training, I worked on an AI-based WhatsApp assistant for a restaurant using n8n, Google Gemini, WhatsApp APIs, and Google Sheets. The system could handle customer questions and food orders while maintaining conversational context. This experience introduced me to the practical side of AI systems and showed me how language models can be integrated with external tools and real-world workflows.

At the same time, I have been actively improving my problem-solving ability through competitive programming. I have solved more than 400 Data Structures and Algorithms problems across platforms such as LeetCode, Codeforces, and SmartInterviews and regularly participate in coding contests. This practice has improved the way I approach unfamiliar problems. Instead of immediately searching for an implementation, I have learned to break problems into smaller parts, analyze constraints, compare approaches, and think about time and space complexity. My ongoing DSA training has also strengthened my interest in algorithms as a fundamental part of computer science.

Beyond technical skills, my academic and project experiences have helped me develop communication, teamwork, and organizational abilities. Working on projects requires discussing ideas, dividing responsibilities, debugging together, and accepting that the first solution is rarely the best solution. I have also learned to independently learn technologies when a project demands them. This ability to learn continuously is one of the qualities I want to carry into graduate study.

I am particularly interested in pursuing graduate study at the University of Toronto because of its strong combination of advanced coursework and research. The MSc in Computer Science is research-oriented and includes a major research project under faculty guidance, which matches my goal of moving beyond application development towards deeper technical investigation. The department's research spans areas such as Artificial Intelligence, Machine Learning, Natural Language Processing, Data Management, Computer Systems, Software Engineering, and Algorithms. These areas closely connect with the interests I have developed through my projects and problem-solving experience. The opportunity to study in an environment where different areas of computer science interact is especially appealing to me.

During my Master's, I want to strengthen my understanding of algorithms, machine learning, intelligent systems, and software systems while developing the ability to conduct independent research. In the short term, my goal is to work on technically challenging projects where I can combine software engineering with AI and data-driven methods. In the long term, I want to become a strong software and AI engineer capable of designing reliable intelligent systems and contributing to research-driven technological development. I am also open to pursuing research further if my graduate experience leads me towards a specific problem worth investigating in depth.

Pursuing a Master's degree abroad is therefore not simply my next academic qualification. It is the next stage of a journey that has already taken me from programming fundamentals to software projects, competitive programming, and practical AI systems. I believe graduate study will provide the depth, research exposure, and challenging environment required for me to grow further. I am prepared to approach the program with the same consistency and curiosity that have shaped my undergraduate journey, and I hope to contribute as an active learner, engineer, and researcher to the university community.